

CAPACITOR CHARGING AND DISCHARGING (10 - A)**Aim**

To determine the capacitance and internal resistance of a given capacitor.

Materials and Apparatus

2 V power supply, electrolytic capacitor, 33 k Ω resistor, multimeter, stopwatch, two-way switch, and leads with crocodile clips.

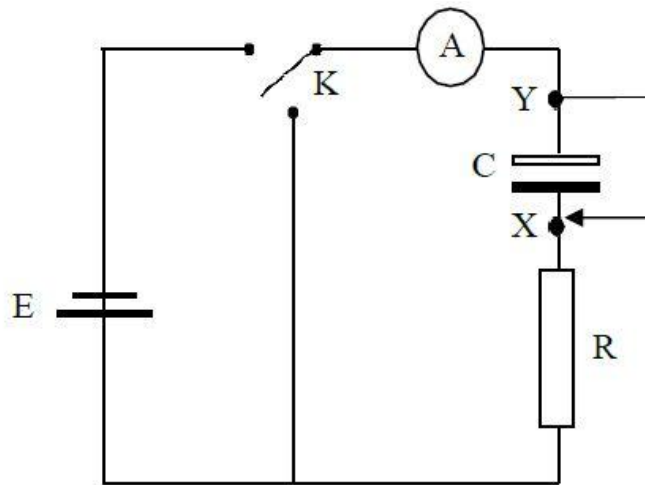
Theory

Fig. 1: The circuit diagram to determine the capacitance and internal resistance. Where, C: Capacitor, R: 33 k Ω resistor, A: Ammeter, K: Two-way switch, E: Power supply.

When the capacitor is charged, the current through the circuit,

$$I = \frac{dq}{dt} \quad (1)$$

And the charge of the capacitor at time t ,

$$q = q_0 \left(1 - e^{-\frac{t}{RC}} \right) \quad (2)$$

Where,

q : Charge of the capacitor

q_0 : Initial charge of the capacitor

t : Time

R : Resistance of the resistor

C : Capacitance of the capacitor

Also, when the voltage (V) is applied, the capacitance (C) of the capacitor is given by,

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$$C = \frac{q}{V} \quad (3)$$

When the capacitor is discharged across a resistor R , the current through the circuit is,

$$I = I_0 e^{-\frac{t}{RC}} \quad (4)$$

Where,

I_0 : Initial current through the circuit

The time constant, $RC = \tau$

From equation (4),

$$\ln\left(\frac{I_0}{I}\right) = \frac{t}{RC} \quad (5)$$

Then, using the graph of $\ln\left(\frac{I_0}{I}\right)$ vs. t , Capacitance can be found.

Procedure

- a) Use the multimeter to measure the power supply voltage.
- b) Set up the circuit as shown in Fig. 1 using the capacitor, marked 'C', taking care to connect the capacitor terminals correctly. The "flying lead" should be connected permanently at Y and the other end should be used only to discharge the capacitor by contacting at X.
- Q) Why should the capacitor be discharged before it is used?
- c) Switch on the charging circuit, connect the flying lead to point X and record the current I_0 of the circuit.
- Q) How does the voltage across the capacitor vary with time when it is charging and discharging?
- Q) Why does the current through the circuit become zero after some time?
- d) Disconnect the flying lead from point X (at the same time start the stopwatch) and record the current I as a function of the time t for a period of 600 s. Switch off the circuit after taking the readings.
- e) Plot the graph of I against t .
- f) Use the graph to estimate the total charge q stored in the capacitor.
- g) Use the above data to determine the capacitance C of the capacitor.
- h) Now switch off the charging circuit and switch on the discharging circuit and record the current I as a function of the time t for a period of 600 s. Switch off the circuit after taking the readings.
- i) Plot the graph of $\ln\left(\frac{I_0}{I}\right)$ vs. t .
- j) Hence determine the internal resistance of the capacitor.

CAPACITOR CHARGING AND DISCHARGING (10 - B)**Aims**

- A. To measure the experimental time constant of a resistor–capacitor circuit.
- B. To compare the time constant to the value from the component values of the resistance and capacitance.
- C. To measure the potential across a capacitor as a function of time as it discharges.
- D. To fit an exponential function to data.

Material and Apparatus

Power Macintosh/Windows PC, 10 μF capacitor, Lab Pro or Universal Lab interface, resistance box (33 $\text{k}\Omega$), Logger Pro, 5V power supply and connecting wires.

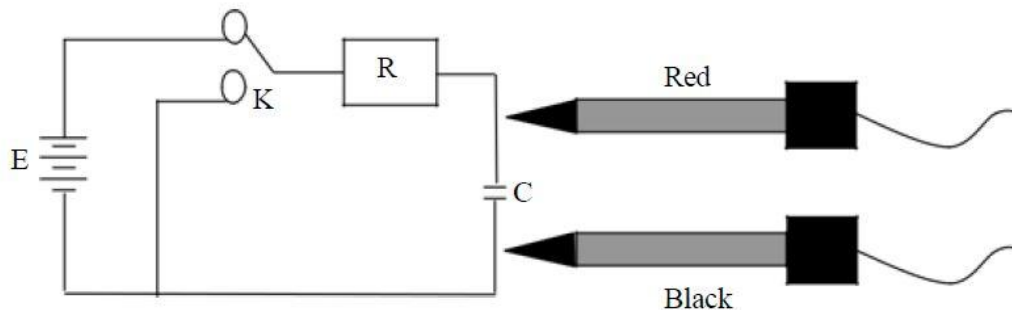
Theory

Fig. 2: Circuit diagram

If an initially charged capacitor of capacitance C to a potential V_0 is discharged across a resistor of resistance R , a time-dependent current flows according to Ohm's law. As the current flows, the charge q is depleted, reducing the potential across the capacitor, which in turn reduces the current, modulated by,

$$V(t) = V_0 e^{-\frac{t}{RC}} \quad (6)$$

For the charging of the capacitor, the potential across it $V(t)$ varies according to,

$$V(t) = V_0 \left(1 - e^{-\frac{t}{RC}} \right) \quad (7)$$

Where,

V_0 : Initial electrical potential across the capacitor

t : Time

R : Resistance of the resistor

C : Capacitance of the capacitor

CAPACITOR CHARGING AND DISCHARGING (10 - B)

Procedure

- a) Connect the circuit as shown in Fig. 1 with the capacitor and the $33\text{ k}\Omega$ resistor.
- b) Connect the voltage probe to channel of the Lab Pro interface, as well as across the capacitor with the red (positive) and the black leads to the capacitor as shown in Fig. 2.
- c) Charge the capacitor until the potential which can be seen at the bottom of the screen, is constant.
- d) Click the “collect” button to begin the data collection. As soon as graphing starts, throw the switch to its other position to discharge the capacitor. Now you can see a plot of the function of potential across the capacitor.
- e) To compare your data to the model, select only the data after the potential has started to decrease dragging across the graph. Click the curve fit tool, select the natural exponential function $Ae^{-C}+B$ and choose a B value and click the button “tryfit”. Repeat the above procedure until you get a nicely fitted function. Record the fit parameters (A, B, C) in your data table. Save your latest run.
- f) Using the fit parameters (A, B, C) determine the experimental time constant and compare that with theoretical time constant.